TRC-OPA1 1 MHz Rail-to-Rail CMOS Operational Amplifier

This is a fictional part, made up for the trace sample project. It is not a real product and no manufacturer publishes it. Do not order it.

1 Features

Single supply 1.8 V to 5.5 V. Gain-bandwidth product 1.0 MHz. Slew rate 0.6 V/us. Supply current 100 uA. Rail-to-rail input and output. Unity-gain stable with 76 degrees of phase margin. SOT-23-5 package.

2 Applications

ADC drivers and buffers, sensor front ends, battery-powered instruments.

3 Pin Configuration

Pin	Name	Type	Description
1	OUT	O	Output
2	V-	P	Negative supply
3	+IN	I	Non-inverting input
4	-IN	I	Inverting input
5	V+	P	Positive supply

6 Electrical Characteristics

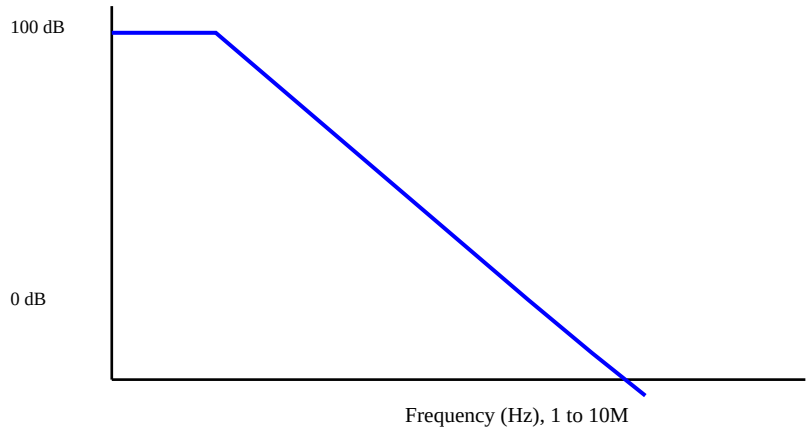
VS = 3.3 V, VCM = VS/2, RL = 1 Mohm to VS/2, TA = 25 C unless noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Supply voltage	VS		1.8		5.5	V
Quiescent current	IQ	no load		100	140	µA
Open-loop voltage gain	AOL	RL = 1 Mohm	90	100		dB
Gain-bandwidth product	GBW		0.8	1.0		MHz
Phase margin	PM	G = +1, CL = 10 pF		76		deg
Slew rate	SR	G = +1	0.45	0.6		V/µs
Common-mode rejection ratio	CMRR	VCM = 0 to VS	62	70		dB
Power-supply rejection ratio	PSRR	VS = 1.8 to 5.5 V	62	70		dB
Input offset voltage	VOS		-4.5		4.5	mV
Input voltage noise density	en	f = 1 kHz		28		nV/√Hz
Output swing from rail	VO	RL = 1 Mohm		25	35	mV
Open-loop output resistance	RO			50		Ω

7 Thermal Information

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Junction-to-ambient thermal resistance	θJA	JEDEC 2s2p		220		°C/W
Junction-to-case thermal resistance	θJC			110		°C/W
Maximum junction temperature	TJ(max)				150	°C

Figure 4. Open-Loop Gain vs Frequency



8 Application: ADC Buffer

As a voltage follower driving a SAR ADC input, place a 100 nF bypass capacitor within 3 mm of V+. The follower settles a 2 V step to 1 % in about 3.5 us.